DA6360 : WS 2

1. $a_* = \max\{\alpha_1, \alpha_2, \dots, \alpha_m\}$
↳ max. of m real numbers α_i ; $i = 1 \dots m$

$$\min_t \; t$$
$$\text{s.t. } t \geq \alpha_i \qquad i = 1 \dots m$$

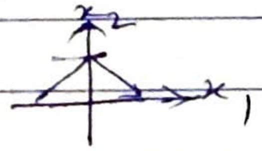
$t \in \mathbb{R}$ (1 var.)
m linear inequalities $t \geq \alpha_i$
obj. min. t
any feasible t must be at least every α_i, therefore, it must be at least $\max_i \alpha_i$. When we minimize t forces it down to smallest value that still satisfies all inequalities.

$$t = \max(\alpha_i, \dots, \alpha_m)$$

So, $\alpha_* = t$

2. A feasible direcⁿ d at a pt. x in a set C means, $\exists$ some $\epsilon > 0$, s.t.

$\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $\qquad x + td \in C \quad \forall\, t \in [0, \epsilon]$

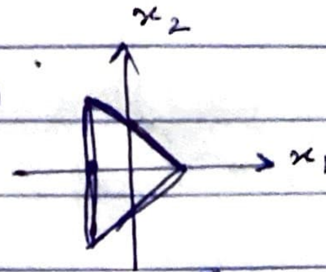
(1) 

$x = (0, 1)$

$d = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\rightarrow (0, 1 + t) \rightarrow$ not feasible

$d = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $\rightarrow (-t, 1) \rightarrow$ not feasible

(2)

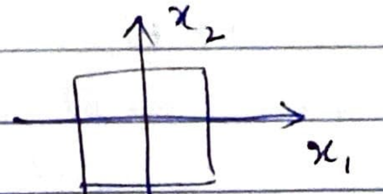


$(-1, 0)$

$d = \begin{bmatrix} 0 \\ 1 \end{bmatrix} : (-1, t) \rightarrow$ feasible

$d \begin{bmatrix} -1 \\ 0 \end{bmatrix} : (-1 - t, 0) \rightarrow$ not feasible

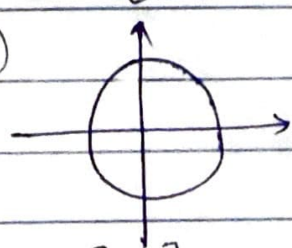
(3)



$x = (0, 1)$

$d = \begin{bmatrix} 0 \\ 1 \end{bmatrix} : (0, 1 + t) \rightarrow$ not feasible

(4)

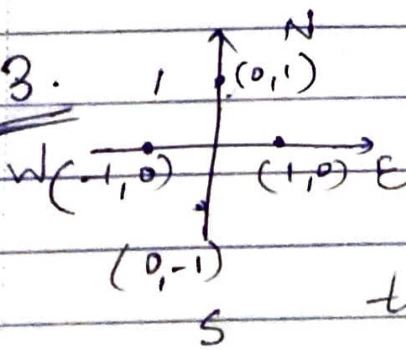


$x = (1, 0)$

$d = \begin{bmatrix} 0 \\ 1 \end{bmatrix} : (1, t) \rightarrow$ not feasible

$d = \begin{bmatrix} -1 \\ 0 \end{bmatrix} : (1 - t, 0) \rightarrow$ ~~not~~ feasible

3.



unconstrained minimisa^n, a direc^n d is improving if the directional derivative is 0.

$$f'(x; d) = \nabla f(x)^T d < 0$$

① $f = (x_1 + x_2)$, $\nabla f = (1, 1)$ (same)

improving direcⁿ → S, W

② $f = -x_1 + x_2$ $\nabla f = (-1, 1)$

improving direcⁿ → E, S

③ $f = x_1 - x_2$ $\nabla f = (1, -1)$

improving direcⁿ → N, W

④ $f = -x_1 - x_2$ $\nabla f = (-1, -1)$

improving direcⁿ → N, E

⑤ $f = 2x_1 + x_2$ $\nabla f = (2, 1)$

improving direcⁿ → S, W

⑥ $f = (x_1 - 1)^2 + (x_2 - 2)^2$

$\nabla f = (2(x_1 - 1), 2(x_2 - 2))$

$(0, 1) \Rightarrow (-2, -2) \Rightarrow$ N, E

$(1, 0) \Rightarrow (0, -4) \Rightarrow$ N

$(0, -1) \Rightarrow (-2, -6) \Rightarrow$ N, E

$(-1, 0) \Rightarrow (-4, -4) \Rightarrow$ N, E

⑦ $f = (x_1 + 3)^2 + (x_2 + 2)^2$

$\nabla f = (2(x_1 + 3) + 2(x_2 + 2))$

$(0, 1) \Rightarrow (6, 6) \Rightarrow$ S, W

$(1, 0) \Rightarrow (8, 4) \Rightarrow$ S, W

$(0, -1) \Rightarrow (6, 2) \Rightarrow$ S, W

$(-1, 0) \Rightarrow (4, 4) \Rightarrow$ S, W

4. $n = 1, 2 \ldots$ → does the following sequences converge

① $\frac{1}{n} \to 0$ (converges)

② $\frac{(-1)^n}{n} \to 0$ (converges)

③ $(-1)^n \to$ oscillates btw. -1 & 1 (DNC)

④ $\sin((-1)^n) = \begin{cases} \sin(1) & \text{odd} \\ \sin(-1) = -\sin(1) & \text{even} \end{cases}$

(DNC)

⑤ $(\sin(-1))^n = (-\sin 1)^n \to 0$ (converge)

⑥ $n^2 \to \infty$ (DNC)

⑦ $\frac{n^2}{2^n} \to 0$ (converges)

⑧ $n^2 (\sin(-1))^n \to 0$ (converges)

⑨ $n^2 (\sin(-0.5))^n \to 0$ (converges)

⑩ $2^n (\sin(-0.5))^n \to 0$ (converges)

5. Find $f'(x)$, $f''(x)$, $f'''(x)$

① $f(x) = x$, $f'(x) = 1$

$f''(x) = 0$, $f'''(x) = 0$

② $f(x) = |x| = \begin{cases} +x & x > 0 \\ -x & x < 0 \end{cases}$

$f'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$ DNE on 0.

$x \neq 0$, $f''(x) = f'''(x) = 0$

③ $f(x) = x^2$, $f'(x) = 2x$

$f''(x) = 2$, $f'''(x) = 0$

④ $f(x) = |x|^2 = x^2$

$f'(x) = 2x$, $f''(x) = 2$, $f'''(x) = 0$

⑤ $f(x) = x^3$, $f'(x) = 3x^2$

$f''(x) = 6x$, $f'''(x) = 6$

⑥ $f(x) = |x|^3 = \begin{cases} x^3 & x > 0 \\ -x^3 & x < 0 \end{cases}$

$f'(x) = \begin{cases} 3x^2 & x > 0 \\ -3x^2 & x < 0 \end{cases}$ $\quad x = 0$, $f'(x) = 0$

$f''(x) = \begin{cases} 6x & x > 0 \\ -6x & x < 0 \end{cases}$ $\quad x = 0$, $f''(x) = 0$

$f'''(x) = \begin{cases} 6 & x > 0 \\ -6 & x < 0 \end{cases}$ $\quad x = 0$, $f'''(x) = 0$

(piecewise)